Project Planning Phase

Sprint Planning Template (Backlog, Sprint Schedule, Stories, Estimation) - Book a Doctor

Date	04 Aug 2026
Student Name	Navendhiran C
Project Name	Book a Doctor (MERN Stack)
Maximum Marks	5 Marks

Note: Date and Team ID are placeholders - update with your actual internship/team details.

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (E	pUSc)N	User Story / Task	Story Point	Priority	Team Member
Sprint-1	**Project Setup & Architecture**	USN-1	As a team, we can draft the ER diagram covering Users, Doctors, and Appointments.	2	High	Deepan
Sprint-1		USN-2	As a developer, I can set up the project folder structure for client and server.	1	High	Deepan
Sprint-1		USN-3	As a developer, I can bootstrap the Node/Express server with an initial route.	2	High	Deepan
Sprint-1		USN-4	As a developer, I can configure the MongoDB connection using Mongoose.	2	High	Deepan
Sprint-1		USN-5	As a developer, I can build Mongoose schemas for User, Doctor, and Appointment.	3	High	Deepan
Sprint-1		USN-6	As a developer, I can set up the React app with routing and Bootstrap styling.	2	Medium	Deepan
Sprint-1		USN-7	As a user, I can sign up for the application by providing email, password, and confirming it.	3	High	Deepan
Sprint-1		USN-8	As a user, I can log into the application by entering email & password.	2	High	Deepan
Sprint-1		USN-9	As a developer, I can build JWT-based authentication middleware.	3	High	Deepan
Sprint-2	**Doctor Discovery & Profiles**	USN-10	As a patient, I can look up doctors by specialization and location.	3	High	Deepan
Sprint-2		USN-11	As a patient, I can view a doctor's profile with experience and ratings.	2	High	Deepan
Sprint-2		USN-12	As a doctor, I can create and edit my profile.	3	High	Deepan
Sprint-2		USN-13	As an admin, I can review and approve new doctor registrations.	3	High	Deepan
Sprint-2		USN-14	As a patient, I can filter/sort doctors by rating or fee.	2	Medium	Deepan

Sprint-2		USN-15	As a developer, I can implement role-based access control for patient/doctor/admin.	3	High	Deepan

Sprint	Functional Requirement (E	pUiSc)N	User Story / Task	Story Point:	Priority	Team Member
Sprint-2		USN-16	As a developer, I can hash passwords using bcrypt.	2	High	Deepan
Sprint-2		USN-17	As a user, I can update my profile information.	2	Medium	Deepan
Sprint-3	**Appointment Booking**	USN-18	As a patient, I can check a doctor's open time slots.	3	High	Deepan
Sprint-3		USN-19	As a patient, I can book an appointment with a doctor.	3	High	Deepan
Sprint-3		USN-20	As a patient, I can reschedule or cancel an existing appointment.	2	Medium	Deepan
Sprint-3		USN-21	As a doctor, I can view and manage my appointment schedule.	3	High	Deepan
Sprint-3		USN-22	As a patient, I can review my appointment history and current status.	2	Medium	Deepan
Sprint-3		USN-23	As a developer, I can add real-time updates for doctor availability.	3	Medium	Deepan
Sprint-3		USN-24	As a patient, I can get an email/SMS confirmation once a booking is made.	2	Medium	Deepan
Sprint-3		USN-25	As a developer, I can test the appointment APIs using Postman.	2	Low	Deepan
Sprint-4	**Documents, Admin & Deployment**	USN-26	As a patient, I can securely upload medical reports and prescriptions.	3	High	Deepan
Sprint-4		USN-27	As a doctor, I can view documents uploaded by my patients.	2	Medium	Deepan
Sprint-4		USN-28	As an admin, I can monitor platform activity and manage user accounts.	3	Medium	Deepan
Sprint-4		USN-29	As a developer, I can tune MongoDB queries for better performance.	2	Low	Deepan
Sprint-4		USN-30	As a developer, I can make sure the UI is responsive across devices.	3	Medium	Deepan
Sprint-4		USN-31	As a developer, I can deploy the application and put together the demo video.	3	High	Deepan
Sprint-4		USN-32	As a developer, I can run the final UAT pass and fix any reported bugs.	4	High	Deepan

Project Tracker, Velocity & Burndown Chart (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Dat	eS)print Release Date (Actu
Sprint-1	20	6 Days	5 Jul 2026	11 Jul 2026	20	4 Aug 2026
Sprint-2	20	6 Days	12 Jul 2026	17 Jul 2026		
Sprint-3	20	6 Days	22 Jul 2026	27 Jul 2026		
Sprint-4	20	6 Days	1 Aug 2026	7 Aug 2026		

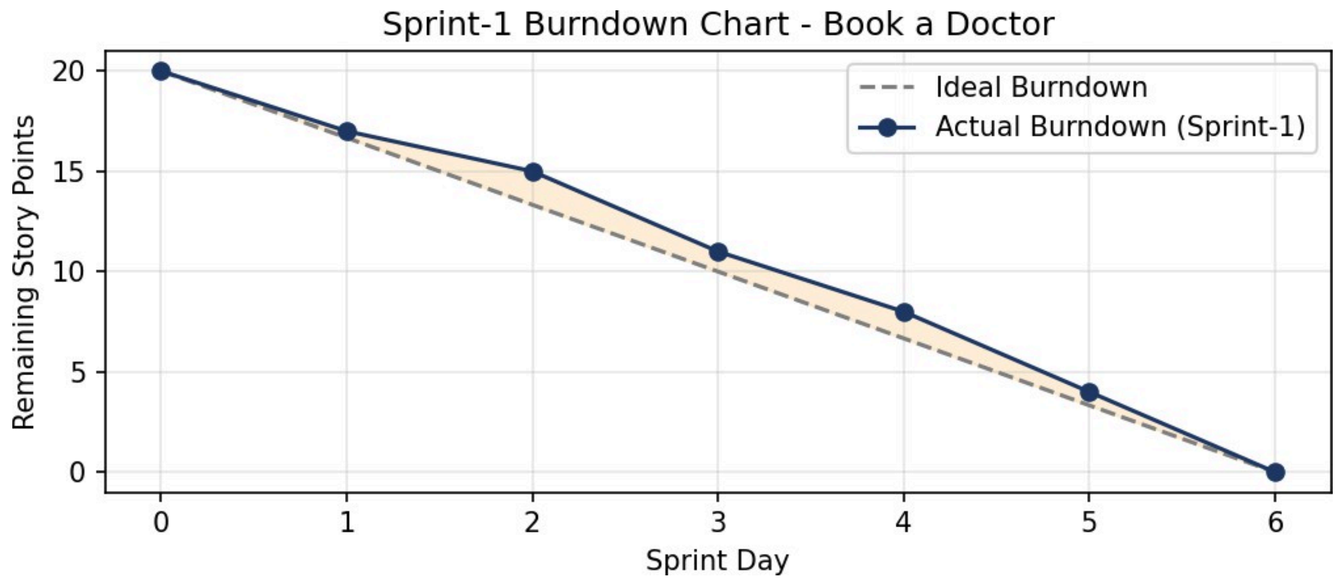
Note: Sprint-1 is completed as a worked example. Fill in Story Points Completed and Actual Release Date for Sprints 2-4 as the team makes progress.

Velocity

Each sprint spans 6 days with a target of 20 story points delivered per sprint. Expressing the team's average velocity (AV) in story points per day: AV = Velocity div Sprint Duration = 20 div 6 roughly 3.3 story points completed per day on average.

Burndown Chart

A burndown chart plots remaining work against time and is a standard tool in Scrum-based agile delivery. The chart below shows Sprint-1 for the Book a Doctor project, contrasting the ideal line with the team's actual progress.



References: visual-paradigm.com/scrum/scrum-burndown-chart | atlassian.com/agile/tutorials/burndown-charts | atlassian.com/agile/project-management | atlassian.com/agile/tutorials/epics |

atlassian.com/agile/tutorials/sprints | atlassian.com/agile/project-management/estimation